

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Cryptography**

Code No: **CSC327** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Illustrate the concept of security policy and mechanism with an example. Differentiate between block cipher and stream cipher. Explain the process of key expansion in AES.
2. Describe the properties of hash functions. Discuss how hash value is generated using SHA-1 algorithm.
3. Show that Z_{25} is a field. John publishes the ElGamal public key $(q, \alpha, Y_a) = (101, 2, 14)$. Jane desired to send the secret message "CSIT" to John. Using the equivalence $A:0, B:1, \dots, Z:25$, encrypt the message using John's public key. Use a random integer $k = 4$.

Section B

Attempt any Eight questions [8×5=40]

4. Differentiate between Trojan horse and virus. Describe any two types of intruders.
5. The message "IMOGUN" was encrypted with a Playfair cipher using keyword "GALOIS". Decrypt the message.
6. How encryption is done using IDEA algorithm.
7. Describe the services provided by Pretty Good Privacy protocol to secure email.
8. What is the condition for two integers, x and y , to be relatively prime? Find whether 61 is prime or not using Miller-Rabin algorithm.
9. Define challenge response system. Why do we need Kerberos?
10. How direct digital signature is different from arbitrated digital signature? How digital signature generation and verification is done using RSA.
11. Which one is more secure, monoalphabetic cipher or poly alphabetic cipher? Justify. Using Rail Fence cipher, encrypt the text "LEARNING AND TEACHING ARE DIFFERENT", using 3 as rails.
12. Why do we need discrete logarithm over normal logarithm? Find out whether 3 is primitive root of 7 or not.